PS No.	**36**
Project Title	**Employee Attendance System**
Problem Statement	Build a console application to add employees, mark their daily attendance as present, absent or leave, and view an attendance report for each employee. Data is saved to a file. **Language:** Python 3 **Concepts:** Dictionaries, Lists, Conditions, File Handling **Note:** Employee IDs start from E101. Status codes: P Present, A Absent, L Leave. Dates use YYYY-MM-DD. Attendance % = Present days ÷ marked days × 100. Data is stored in attendance.txt. Test cases run in order on the same records.
Test Cases / Case Statements (4 cases)	

INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
Menu option 1 (Add employee) Name: Aman	Employee added: E101 | Aman
Test Case 2: Boundary / Minimum Input	
Employee E101 has no attendance records Menu option 3 (Report): E101	Aman: Present 0 | Absent 0 | Leave 0 Attendance: 0.00% (no records)
Test Case 3: Invalid Input	
(a) Menu option 2 (Mark): E999, 2026-08-01, P (b) Mark: E101, 2026-13-45, P (c) Mark: E101, 2026-08-01, X	(a) Employee E999 not found. (b) Invalid date. Use YYYY-MM-DD. (c) Invalid status. Enter P, A or L.
Test Case 4: Different Valid Scenario	
Employee E101 Aman Mark: 2026-08-01 P, 2026-08-02 P, 2026-08-03 A, 2026-08-04 L, then 2026-08-02 P again Menu option 3 (Report): E101	Attendance for 2026-08-02 is already marked. Aman: Present 2 | Absent 1 | Leave 1 Attendance: 50.00%

Bare Minimum Requirements	1. Write the solution in Python 3 (one .py file) and run it from the console. 2. Store each employee as a dictionary (id, name, records) inside a list. 3. Write functions to add an employee, mark attendance and show the report. 4. Validate the employee ID, the date format and the status, and block duplicate attendance for the same date. 5. Load the data from the file at start and save it after every change.
Instruction	1. Understand the problem statement and define the problem clearly. 2. Develop the solution using proper logic and programming concepts. 3. Implement all required features and pass the given test cases. 4. Test and debug the project to ensure correct output. 5. Prepare the documentation/report and upload the code and report to GitHub. 6. Complete and submit the project within the given deadline.